

● MEDIUM

● GEOMETRY AND TRIGONOMETRY

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# Right triangles and trigonometry

# 02

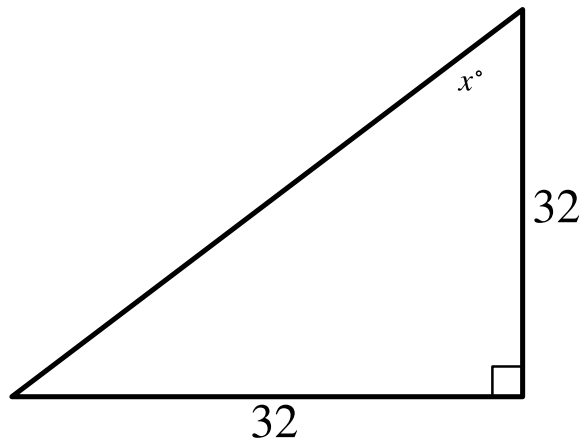
4 questions · ~6 min



Q1

GRID-IN

GEOMETRY AND TRIGONOMETRY



Note: Figure not drawn to scale.

- One angle is a right angle.
- The measure of a second angle is  $x^\circ$ .
- The lengths of the 2 sides adjacent to the right angle are as follows:
  - 32
  - 32
- Note: Figure not drawn to scale.

In the triangle shown, what is the value of  $x$ ?

STUDENT-PRODUCED RESPONSE

|                      |                      |                      |                      |
|----------------------|----------------------|----------------------|----------------------|
| <input type="text"/> | <input type="text"/> | <input type="text"/> | <input type="text"/> |
| .                    | /                    | .                    | .                    |
| 0                    | 0                    | 0                    | 0                    |
| 1                    | 1                    | 1                    | 1                    |
| 2                    | 2                    | 2                    | 2                    |
| 3                    | 3                    | 3                    | 3                    |
| 4                    | 4                    | 4                    | 4                    |
| 5                    | 5                    | 5                    | 5                    |
| 6                    | 6                    | 6                    | 6                    |
| 7                    | 7                    | 7                    | 7                    |
| 8                    | 8                    | 8                    | 8                    |

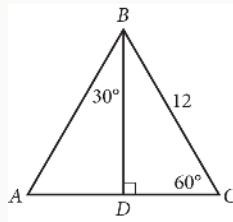
Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

**Q2**

GEOMETRY AND TRIGONOMETRY

Triangle  $FGH$  is similar to triangle  $JKL$ , where angle  $F$  corresponds to angle  $J$  and angles  $G$  and  $K$  are right angles. If  $\sin F = \frac{308}{317}$ , what is the value of  $\sin(J)$ ?

- A.  $\frac{75}{317}$
- B.  $\frac{308}{317}$
- C.  $\frac{317}{308}$
- D.  $\frac{317}{75}$



In  $\triangle ABC$  above, what is the length of  $\overline{AD}$ ?

A. 4

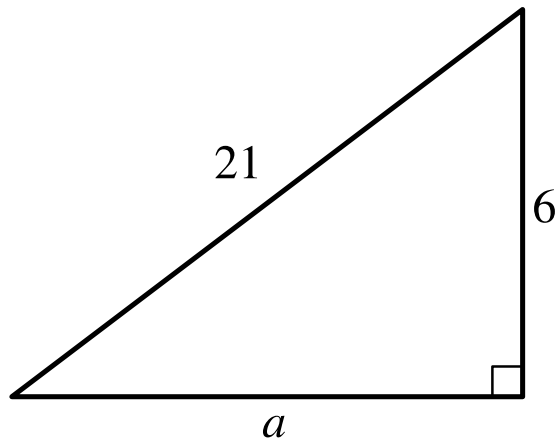
B. 6

C.

$$6\sqrt{2}$$

D.

$$6\sqrt{3}$$



Note: Figure not drawn to scale.

- One angle is a right angle.
- The length of the side opposite the right angle is 21.
- The length of one side adjacent to the right angle is 6.
- The length of the other side adjacent to the right angle is  $a$ .
- A note indicates the figure is not drawn to scale.

For the triangle shown, which expression represents the value of  $a$ ?

- A.  $\sqrt{21^2 - 6^2}$
- B.  $21^2 - 6^2$
- C.  $\sqrt{21 - 6}$
- D.  $21 - 6$